8 Projects (optional, for fun) FULL

Project 8. Your sensor readings on your own website

Quick version

No time to read it all? This is the short version. The full guide is below.

- A small Python script copies your soil sensor's number from Home Assistant into your website's database every 15 minutes, for your **Data** page.
- Hard, about 3 hours, free. Do Projects 3 and 6, and guide 03, first.
- Make a Home Assistant long-lived token and save it in the private file `docker/.ha.env` as `HA_TOKEN=...`.
- Put your sensor's name in `SENSORS` in `docker/extras/ha_to_garden.py`.
- Run it once: `python3 docker/extras/ha_to_garden.py`. You should see `Saved 1 reading(s) at ...`.
- Make it run every 15 minutes with one line in `crontab -e`.
- Finding your number on the Data page is exploring. If it does not show, `rust/junsgarden/src/pages/data.rs` explains which widgets read it.

Level: hard. **Time:** about 3 hours. **Cost:** free.

What you will build

A small Python script that copies your soil sensor's number from Home Assistant into your garden website's database every 15 minutes, so it can show on your website's **Data** page.

This ties your hardware projects to your website.

```
ESP32 sensor --> Home Assistant --script (every 15 min)--> garden database --> Data page
```

What you need first

- Projects 3 and 6 done, and guide 03.
- Python 3 (Ubuntu already has it: check with `python3 --version`).

Steps

Part A. Let the script read Home Assistant

1. In Home Assistant click your **name** (bottom left), **Security** tab, scroll to **Long-lived access tokens**, **Create token**, name it `garden script`. Home Assistant shows it **once**. Copy it.
2. Save it in a private file (this does not print it). In the project folder:

```
umask 077
nano docker/.ha.env
```

Put exactly this one line, with your token after the `=`:

```
HA_TOKEN=paste-the-token-here
```

Save (`Ctrl+O`, `Enter`, `Ctrl+X`). The project's `.gitignore` already keeps `*.env` files out of git.

Part B. Tell the script which sensor

3. Find your sensor's name: HA, **Settings, Devices & Services, Entities**, search `moisture`. Copy the name that starts with `sensor.`.

4. Open `docker/extras/ha_to_garden.py` in an editor. Find `SENSORS`. Put your sensor's name on the right side.
5. Read the script from the top. Every step has a comment. Ask Gemini (prompt 4) to explain any part.

Part C. Run it once

6. Run:

```
python3 docker/extras/ha_to_garden.py
```

You should see: `Saved 1 reading(s) at ...`. 7. Check the database has it:

```
ROOTPW=$(grep '^DB_ROOT_PASSWORD=' .env | cut -d= -f2-)
docker exec -e MYSQL_PWD="$ROOTPW" garden-db mariadb -uroot -e \
  "SELECT obs_time, location, source, soil_moisture_6in FROM juns_garden_db.weather
ORDER BY id DESC LIMIT 3;"
```

You should see: your row at the top.

Part D. Make it run by itself

8. `crontab -e` and add (change `jun`):

```
*/15 * * * * python3 /home/jun/junsgarden-for-jun/docker/extras/ha_to_garden.py >>
/home/jun/junsgarden-for-jun/backups/ha-to-garden.log 2>&1
```

Part E. Find it on the website

9. Open `http://localhost:8030/data.php` (the Data page). Look for a chart that uses soil moisture. *This part is exploring.* The Data page was built for The author's weather data and different widgets read different columns.
10. If your numbers do not appear where you expect, that is a real programming puzzle. Work it out like this:

1. Open `rust/junsgarden/src/pages/data.rs` and search for `soil_moisture_6in`. The comments explain which widgets read it.
2. Ask Gemini (prompt 4) to explain how that widget picks its rows (the `WHERE` part of the SQL).
3. If it filters by a `location` or `source` you did not use, change the two labels at the top of `ha_to_garden.py` to match.

How you know it worked

- A new row appears in the database every 15 minutes.
- Your sensor's number shows on your Data page.

If something goes differently

What you see	What to do
`No HA_TOKEN`	Part A, step 2: the line must be `HA_TOKEN=...` with nothing before it.
`could not read ...: HTTP Error 401`	The token was copied wrong or deleted. Make a new one.
`HTTP Error 404`	The sensor name is wrong. Copy it from Entities.
`No such container: garden-db`	`docker compose up -d` first.
Cron seems to do nothing	Look at `backups/ha-to-garden.log`.

Make it yours

- Add the air temperature and humidity (uncomment the two example lines).
- Add a new widget to the Data page. Start by reading how an existing one is written in `data.rs`. Make the smallest possible copy and change one word.

- Write a README for your script (3 sentences) for your MY-WINS file.

Next: Project 9: More ideas

loved.
endured.
never forgotten.